

RISHABH GOEL, *Biomedical Data Scientist*

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PROFESSIONAL PROFILE

Biomedical data scientist and entrepreneur with 3+ years of industry experience in leveraging multi-modal data analysis with clinical and genomic data, machine learning techniques, and state-of-the-art deep learning tools to drive innovative solutions in healthcare.

EDUCATION

HARVARD MEDICAL SCHOOL

MAY 2026

Masters of Medical Sciences in Biomedical Informatics, GPA: 4.0

Boston, MA

- Coursework: Advanced Coding and Statistics for Biomedical Informatics; Precision Medicine: Integrating Clinical and Genomic Data; Foundations of Biomedical Informatics; Revolutionary Ventures: How to Invent and Deploy Transformative Technologies

UNIVERSITY OF CALIFORNIA, BERKELEY

MAY 2022

Bachelor of Arts in Molecular and Cellular Biology: Biochemistry (Minor in Data Science), GPA: 3.68

Berkeley, CA

Bachelor of Science in Business Administration

- Coursework: Biophysical Chemistry; Pathways Mechanisms and Regulation; Advanced Machine Learning Algorithms; Principles & Techniques of Data Science; Structure and Interpretation of Computer Programs; Data Structures and Algorithms

TECHNICAL SKILLS

Technical Languages/Tools: Python, R, Jupyter Notebook, SQL, Matlab, Java, GitHub, AWS (S3, EC2, Athena), GCP, Google Colab

Data Science Libraries: Pandas, Sklearn, SciPy, NumPy, PyTorch, TensorFlow, Matplotlib, BeautifulSoup, Plotly, Seaborn

Bioinformatics Tools/Skills: Bulk RNA-Seq, ScRNA-Seq, Bioconductor, Public Databases (GEO, TCGA, Drugbank, PRIDE, Uniprot)

RESEARCH EXPERIENCE

Computational Research Specialist

July 2023 – July 2024

Dai Lab for Molecular Design and Synthetic Biology, Rice University, Houston, TX

Principal Investigator: Dr. Mingjie Dai

- Automated the sequencing read alignment for raw sequencing data from FASTA files using **Bowtie 2** in **Python** which resulted in a 5-fold increase in efficiency of the data processing pipeline
- Advanced the development of primer design software for cPCR by building a simulated annealing optimization algorithm using **MATLAB** that enables primer design up to 121-plex with minimal primer crosstalk or primer-dimer formation
- Tested the performance of the software on an 86-gene marker T-Cell panel by using **APIs** to extract gene sequences from public genomic databases such as **NCBI Genome Data Viewer** and **UCSC Genome Browser** resulting in **15% improvement of software**

Data Discovery Project Research Fellow

December 2021 – May 2022

Merck & Co, Rahway, NJ

Principal Investigator: Dr. Xiang Yu

- Awarded a Data Science Discovery Program **3-month Research Fellowship with Merck** from the UC Berkeley Computing, Data Science, and Society Department to explore deep learning techniques in pharmacology research
- Designed a model to predict pharmacokinetics (PK) in patients using neural networks (**Neural-ODEs**) with **TensorFlow**, which produced a **4% increase in accuracy** over state-of-the-art techniques
- Organized and **presented project resources** including code, methods, findings, and results on **GitHub** for the accessibility and perusal of key stakeholders which **improved project visibility by 50%**
- Presented my work as a poster at the 2023 Academic Data Science Alliance (ADSA) Annual Meeting in San Antonio, Texas, and won the award for the **"Best Undergraduate Research Project"** at the conference amongst over 50 projects

Biomedical Informatics Research Associate

May 2021 – December 2021

Biomedical Informatics Jegga Lab, Cincinnati Children's Hospital, OH

Principal Investigator: Dr. Anil Goud Jegga

- Developed a tool to accelerate drug repurposing by comparing gene expression changes associated with drug use to changes in disease, extracted from over 1,100 experiments in **Gene Expression Omnibus (GEO)** and **EBI Atlas**
- Validated disease hypotheses by conducting **gene set enrichment analysis (GSEA)** using public pathway databases (**KEGG pathways** and **Gene Ontology (GO)**) which enabled the discovery of 2 novel disease indications for a p38 α MAP kinase inhibitor
- Built visualizations to depict the correlation between the drug and disease **differential gene expression** such by using **R** to create **heatmaps** and **volcano plots** that increased the efficiency of our disease selection process 2-fold

PROFESSIONAL EXPERIENCE

Co-founder

August 2023 – August 2024

Doorstep.ai, San Francisco, CA

- **Raised ~\$1.8 Million in investment** from notable backers including Cercano Management (Co-founder of Microsoft - Paul Allen's Family Office), Antler, Kleiner Perkins Scout (Sean Henry, Founder & CEO of Stord), Cassius VC, Richard Sussman, and others
- Led a team of 3 software engineers to build a software product that enables **100% accurate last-mile delivery through AI-based visual guidance** for drivers to get to the customer's doorstep
- Developed computer vision algorithms using **Convolutional Neural Networks** with **PyTorch** that predict the location of a user down to 0.5 meters of accuracy (for reference today's GPS technologies have 8-fold lower accuracy)
- Built machine learning algorithms using **XGBoost** on motion data from a users phone that predicted the phone's pose (in the pocket, swinging, reading context) with **over 99% accuracy**
- **Awarded ~\$50,000 non-dilutive grant** awards from multiple venture funds and pitch competitions including 1789 Student Venture Fund, Innovate Carolina, TiE University Global Pitch Competition, and LAUNCH, University of California's Startup Accelerator

Biomedical Data Scientist

May 2022 – March 2024

GEn1E Lifesciences, Palo Alto, CA

Principal Investigators: Dr. Ritu Lal & Dr. Mridul Mehta

- Developed a machine learning model to identify Acute Respiratory Distress Syndrome (ARDS) patient subphenotypes using **support vector machine** model that was trained on high-dimensional clinical and biomarker data, which achieved **97%+ accuracy**
- [This work received a \\$749,000 grant from BARDA's Division of Research, Innovation, and Ventures \(DRIVe\) for Patient Stratification Using AI](#). I also **co-authored a journal article** with these findings currently under review at the BMJ.
- Awarded the finalist position at the **Digital Health Hub Awards 2023** for having one of the top health equity digital platforms amongst over 100 other competing companies
- Accepted into **NVIDIA's Inception Program**, a highly-selective invitation to join an elite group of startups driving change across industries and around the world

SELECT AWARDS

- Provisional patent for 'Techniques For Machine Learning-Assisted Precise Wayfinding'
- Provisional patent for 'Machine Learning Systems for Acute Respiratory Distress Syndrome Patient Sub-Phenotyping'
- [\\$749,000 grant from BARDA's Division of Research, Innovation, and Ventures \(DRIVe\) for Patient Stratification Using AI](#)
- \$1.8 Million investment for Doorsep.ai from notable backers including Paul Allen's Family Office, Antler, Kleiner Perkins Scout
- \$50,000 grant awards from multiple venture funds and pitch competitions including TiE University Global Pitch Competition
- *Best Undergraduate Student Research Project* at the Academic Data Science Alliance (ADSA) Annual Meeting, 2023
- Data Science Discovery Program 3-month Research Fellowship with Merck

SELECT PUBLICATIONS AND PRESENTATIONS

- Mehta M, **Goel R**, Goel NN, Satterwhite L, Das M, Zainab A, Reynolds D, Babar I, Gidwani UK, Lal R, Dholakia S. *Development and Validation of a Prospective Machine Learning Model for the Identification of ARDS Sub-phenotypes* [manuscript submitted for review with Manuscript No. CHEST-CRITCARE-D-24-00041]. 2024.
- Lal R, **Goel R**, Mehta M, Goel NN, Satterwhite L, Das M, Zainab A, Reynolds D, Babar I, Gidwani UK, Dholakia S. *Development and Validation of a Prospective Machine Learning Model for the Identification of ARDS Sub-phenotypes* [Presentation at the 2024 ATS International Conference, San Diego]. 2024.
- **Goel, R.** (2023). Is Neural-ODE an Effective Method for Modeling Pharmacokinetics (PK) Data in Real-world Clinical Settings? *Academic Data Science Alliance*. Retrieved from <https://adsa.pubpub.org/pub/d5dohi00>